

# 431 Lab 4 Instructions

Fall 2026 - deadline in [Course Calendar](#)

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## Table of contents

|                                                           |   |
|-----------------------------------------------------------|---|
| 0.1 R Setup .....                                         | 2 |
| 0.2 Getting Help .....                                    | 2 |
| 0.3 Learning Objectives for this Lab .....                | 2 |
| 1 Task 1 (10 points) .....                                | 3 |
| 1.1 The Data .....                                        | 3 |
| 1.2 The Task .....                                        | 3 |
| 2 Task 2 (15 points) .....                                | 4 |
| 3 Task 3 (10 points) .....                                | 4 |
| 3.1 The Data .....                                        | 4 |
| 3.2 The Task .....                                        | 4 |
| 4 Task 4 (15 points) .....                                | 5 |
| 5 Section 5 of your Lab Report: AI Usage .....            | 5 |
| 6 Section 6 of your Lab Report: Session Information ..... | 5 |
| 7 Additional Notes and Instructions .....                 | 5 |
| 7.1 Submitting this Lab .....                             | 5 |
| 7.2 Grading this Lab .....                                | 5 |
| 7.3 Emergencies and Late Policy .....                     | 6 |
| 7.4 Lab Regrade Requests .....                            | 6 |
| 8 Session Information .....                               | 6 |

### ! Important

- This Lab contains 4 tasks for you to complete.
- The deadline for completing this Lab is posted in the [Course Calendar](#).

## 0.1 R Setup

```
library(tidyverse)
```

You will use additional packages in developing your response.

We have built no template for Lab 4. Instead, use the templates for Lab 1 and Lab 2 we have provided to help you get started.

## 0.2 Getting Help

You may discuss each Lab with Professor Love, the teaching assistants or your colleagues, but your answer must be prepared by **you working alone**. Don't be afraid to ask questions, using any of the methods described on [our Contact Us page](#).

## 0.3 Learning Objectives for this Lab

1. Ingest data from an Excel (.xlsx) file into R
2. Complete appropriate analyses of multiple two-way contingency tables.
3. Communicate effectively about your results in complete, clear English sentences.
4. Demonstrate your progress to date on Project A's data management activities.

# 1 Task 1 (10 points)

## 1.1 The Data

For Tasks 1 and 2, we use the data in the `lab4-task1.xlsx` (Excel) file, which is derived from the LicoriceGargle example at the [Cleveland Clinic Statistical Dataset Repository](#) which provides two files to explain the data.

- a [general description](#), and
- a [data description](#).

The data were originally published in Ruetzler K et al. (2013) [A randomized, double-blind comparison of licorice versus sugar-water gargle for prevention of postoperative sore throat and postextubation coughing](#) *Anesth Analg* 117: 614-21. <https://doi.org/10.1213/ane.0b013e318299a650>.

From the Abstract to the Ruetzler paper, we have the following details:

We enrolled 236 patients having elective thoracic surgery who required intubation with a double-lumen endotracheal tube. Patients were randomly assigned to gargle 5 minutes before induction of anesthesia for 1 minute with: (1) Extractum Liquiritiae Fluidum (licorice 0.5 g); or (2) Sirupus Simplex (sugar 5 g); each diluted in 30 mL water.

Our small sample of the original data includes only 235 rows and three variables:

| Variable                  | Description                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>subject</code>      | Subject ID, added by Dr. Love                                                                                  |
| <code>intervention</code> | sugar or licorice (from <code>treat</code> in the CCF dataset)                                                 |
| <code>postop4</code>      | Yes or No regarding sore throat pain at rest 4 hours after surgery (from <code>post0p4hour_throatpain</code> ) |

## 1.2 The Task

Compare the rates of post-operative sore throat pain among subjects who gargled with licorice as compared to those who gargled with sugar. In particular, create and share an appropriate statistical summary of the relationship, and use appropriate statistical methods to estimate the point estimate and 95% confidence intervals for an appropriate:

- relative risk,
- odds ratio, and
- risk difference.

In this case, do not use a Bayesian augmentation.

## 2 Task 2 (15 points)

Reflecting on what you've done in Task 1 and what you've read in Spiegelhalter's *The Art of Statistics*, please write an essay of 125-200 words which discusses the relationship between `postop4` and `intervention` that you observe in this sample.

In your essay, first, describe what you have learned about the association of licorice (vs. sugar) and sore throat pain 4 hours post-surgery from this work, avoiding jargon and also avoiding statements about statistical significance. Then, discuss how what we know about the study design tells us about what we might be able to conclude about whether gargling with licorice reduces pain at 4 hours post-surgery.

## 3 Task 3 (10 points)

### 3.1 The Data

The `lab4-task3.Rds` dataset (also known as the *lindner* data) provided on our [431-data page](#) comes from an observational study of 996 patients receiving an initial Percutaneous Coronary Intervention (PCI) at Ohio Heart Health, Christ Hospital, Cincinnati in 1997 and followed for at least 6 months by the staff of the Lindner Center.

The 698 patients thought to be more severely diseased were assigned to treatment with **abciximab** (an expensive, high-molecular-weight IIb/IIIa cascade blocker); while the remaining 298 patients received **usual care** with their initial PCI. Additional information on the [lindner data set](#) is available [here](#).

These data relate to Kereiakes DJ, Obenchain RL, Barber BL, et al. Abciximab provides cost effective survival advantage in high volume interventional practice. *Am Heart J* 2000; 140: 603-610.

### 3.2 The Task

Ingest the `lab4_task3.Rds` data into R, and use them to develop an appropriate comparison of the relative risk of an `acutemi` for those receiving abciximab compared to those receiving usual care. Be sure to provide your code, and interpret your results in context in at least two English sentences. Use a 90% confidence level.

#### Tips for Task 3

1. You should be changing the variable type and labels to make the results more interpretable (perhaps with `fct_recode()`), as well as change the levels so we are obtaining the probability or odds of a myocardial infarction for those who received abciximab compared to those who received usual care in a contingency table with abciximab status in the rows and acute MI status in the columns.
2. An appropriate contingency table will have the value for subjects who have an acute MI and who are receiving abciximab in the top left, and that cell should contain between 100 and 150 subjects.

## 4 Task 4 (15 points)

Answer these questions about your Project A and its data.

Please write a complete sentence in response to each part of Task 4 that clarifies what you are answering *rather than* including each question in your Lab 4 Quarto file.

- a. Are you working with a partner on your project, and if so, who is it?
- b. Which random seed have you selected to choose your two states?
- c. Which two states did this lead to you selecting in addition to CA, FL, NY and OH?
- d. What are the dimensions (rows and columns) of your final clean analytic tibble, **projA\_master**?
- e. What is the minimum value of the percentage of adults reporting poor or fair health (age-adjusted) in your **projA\_master** data set?
- f. What is the number of counties with each possible value of `water_v` in your **projA\_master** tibble?
- g. What is the minimum value of the `lbw_old` variable in your **projA\_master** tibble?
- h. How many lines long is your current Project A Quarto file? (Note that we expect your answer to be at least 20 lines. If this isn't true, add that Quarto file to your submission.)

## 5 Section 5 of your Lab Report: AI Usage

All students should include an AI Usage section in each assignment for this class. See the instructions from Lab 1 for more details.

## 6 Section 6 of your Lab Report: Session Information

Include the session information as a final section in this Lab. I've done so at the bottom of this document.

## 7 Additional Notes and Instructions

### 7.1 Submitting this Lab

Submit this Lab via [Canvas](#), using the Lab 1 assignment. Be sure to submit all three files:

1. Your Quarto file (.qmd) built using our Lab 1 template.
2. The HTML file you obtain by knitting the Quarto file (.html)
3. Your video file (.mp4 or .mov preferred)

Be sure that your Quarto (and thus HTML) files include the **AI Help** and **Session information** sections at the end of the document, and that these section headings appear in the Table of Contents.

### 7.2 Grading this Lab

This Lab will be graded by the TAs and then reviewed by Dr. Love. Your grades will be available one week after the Lab deadline.

The maximum score on this Lab is 50 points.

As each Lab passes its deadline (as listed in the [Course Calendar](#)), we will:

- post the answer sketch (48 hours after the deadline) and draft grading rubric to our Shared Google Drive, and then
- post grades and any revisions to the grading rubric or answer sketch one week after the deadline to a location we will provide to you.

### 7.3 Emergencies and Late Policy

We do not grant extensions on Lab deadlines.

- To receive full credit on a Lab, it must be received on Canvas no later than 59 minutes after the posted deadline. (This allows for small issues with uploading to Canvas to occur without penalty.)
  - Labs that are turned in 1-48 hours after the deadline will lose 10 points for late work.
- No extensions to Lab deadlines will be made this semester. Labs turned in more than 48 hours after the deadline will receive no credit, since by then the Lab Sketch will be posted.
- For details on how Lab grades affect your course grade, see the Syllabus.
- If you have an emergency that will keep you from submitting the Lab by even the late deadline of Friday at noon, please let Dr. Love know that (as soon as possible) via email and he will consider excusing you from the Lab.

### 7.4 Lab Regrade Requests

If, after your Lab is graded, you want Dr. Love to review the grading or correct a grading error, please follow the Lab Regrade Request policy [posted on our Labs page](#).

## 8 Session Information

```
xfun::session_info()
```

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Locale:
```

```
LC_COLLATE=English_United States.utf8
LC_CTYPE=English_United States.utf8
LC_MONETARY=English_United States.utf8
LC_NUMERIC=C
LC_TIME=English_United States.utf8
```

```
Package version:
```

```
askpass_1.2.1      backports_1.5.1    base64enc_0.1.6
bit_4.6.0          bit64_4.8.2        blob_1.3.0
broom_1.0.13       bslib_0.11.0       cachem_1.1.0
```

|                   |                    |                     |
|-------------------|--------------------|---------------------|
| callr_3.8.0       | cellranger_1.1.0   | cli_3.6.6           |
| clipr_0.8.1       | compiler_4.6.1     | conflicted_1.2.0    |
| cpp11_0.5.5       | crayon_1.5.3       | curl_7.1.0          |
| data.table_1.18.4 | DBI_1.3.0          | dbplyr_2.6.0        |
| digest_0.6.39     | dplyr_1.2.1        | dtplyr_1.3.3        |
| evaluate_1.0.5    | farver_2.1.2       | fastmap_1.2.0       |
| fontawesome_0.5.3 | forcats_1.0.1      | fs_2.1.0            |
| gargle_1.6.1      | generics_0.1.4     | ggplot2_4.0.3       |
| glue_1.8.1        | googledrive_2.1.2  | googlesheets4_1.1.2 |
| graphics_4.6.1    | grDevices_4.6.1    | grid_4.6.1          |
| gtable_0.3.6      | haven_2.5.5        | highr_0.12          |
| hms_1.1.4         | htmltools_0.5.9    | httr_1.4.8          |
| ids_1.0.1         | isoband_0.3.0      | jquerylib_0.1.4     |
| jsonlite_2.0.0    | knitr_1.51         | labeling_0.4.3      |
| lifecycle_1.0.5   | lubridate_1.9.5    | magrittr_2.0.5      |
| memoise_2.0.1     | methods_4.6.1      | mime_0.13           |
| modelr_0.1.11     | openssl_2.4.2      | otel_0.2.0          |
| pillar_1.11.1     | pkgconfig_2.0.3    | prettyunits_1.2.0   |
| processx_3.9.0    | progress_1.2.3     | ps_1.9.3            |
| purrr_1.2.2       | R6_2.6.1           | ragg_1.5.2          |
| rappdirs_0.3.4    | RColorBrewer_1.1-3 | readr_2.2.0         |
| readxl_1.5.0      | rematch_2.0.0      | rematch2_2.1.2      |
| reprex_2.1.1      | rlang_1.3.0        | rmarkdown_2.31      |
| rstudioapi_0.19.0 | rvest_1.0.5        | S7_0.2.2            |
| sass_0.4.10       | scales_1.4.0       | selectr_0.6.0       |
| stats_4.6.1       | stringi_1.8.7      | stringr_1.6.0       |
| sys_3.4.3         | systemfonts_1.3.2  | textshaping_1.0.5   |
| tibble_3.3.1      | tidyr_1.3.2        | tidyselect_1.2.1    |
| tidyverse_2.0.0   | timechange_0.4.0   | tinytex_0.60        |
| tools_4.6.1       | tzdb_0.5.0         | utf8_1.2.6          |
| utils_4.6.1       | uuid_1.2.2         | vctrs_0.7.3         |
| viridisLite_0.4.3 | vroom_1.7.1        | withr_3.0.3         |
| xfun_0.60         | xml2_1.6.0         | yaml_2.3.12         |